PAA platform research and plan

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Bottom line

Keep your iQOO Z7 Pro stock and locked, run PAA as a sideloaded app on it, and add cloud telephony for AI calls: it is the only path I found that can keep all four of your hard constraints without tricks, provided Axis passes Tests 1 and 11. Everything below is tagged **[Sourced]**, **[Inference]** or **[Unverified, test it]**.

- **Bootloader:** I found no working legitimate unlock for the iQOO Z7 Pro or any iQOO Z-series phone. The "OEM unlocking" toggle is reported to be cosmetic on vivo/iQOO. Re-locking with custom keys: no evidence it exists. **[Sourced, but weak sources: forum threads and one blog; absence of evidence is not proof.]**
- **Custom OS with verified boot:** only Pixels are supported by GrapheneOS because they allow flashing your own verified-boot key and locking again. Indian reports are mixed: Paytm is reported blocked on GrapheneOS (Feb 2026); PhonePe and the new HDFC app reportedly work; I found no solid 2026 report for Axis Mobile, BHIM or Google Pay. **[Sourced, user reports.]**
- **AI talking on a normal SIM call:** I found no legitimate way for an app to inject speech into a cellular call on Android. Use a cloud number (Exotel, Plivo, Twilio) for AI-voice calls, and expect TRAI rules for automated calls (new rules dated 18 Sep 2026). **[Sourced + Inference.]**
- **Banking data:** an individual cannot be an Account Aggregator data user; the regulated-entity requirement blocks it. Use SMS alerts, notifications and statements instead. **[Sourced.]**
- **Biggest risk to your plan:** Indian bank/UPI apps reportedly warn or block when USB/wireless debugging, Shizuku or a third-party accessibility service is on (ICICI documents it; users report BHIM, and a May 2026 forum post reports Axis refusing to run when another app has accessibility). Your own PAA uses accessibility, so this must be tested first (Test 1). **[Sourced: bank notice for ICICI, user reports for Axis and BHIM; Test 1 and Test 11 settle it.]**

Platform paths compared

Five of the six platform paths keep verified boot, but only A and B run on the phone you own today (G is an add-on). Ratings combine sources and my inference, marked in the last column.

Path	Agent power	Security (verified boot, attestation)	Banking / UPI	Effort	Risks	Evidence
A. Stock iQOO, locked, sideloaded PAA (accessibility, notification listener, SMS via restricted-settings grant, cloud telephony)	Medium: UI automation + notifications; no call audio, no privileged APIs	Full vendor verified boot; Play Integrity device/strong available	Best available; bank apps may still warn on accessibility service (BHIM reported)	Low	Per-app accessibility detection; Android 15 restricted-settings friction; sideload rules tightening	Sourced + Inference
B. Path A + always-on Shizuku (wireless debugging)	High: input injection, screencap, grant permissions, am start; call recording possible on some phones	Bootloader stays locked, but developer options/ADB stay on	Worse: ICICI documents blocking with USB debugging on; BHIM reported flagging Shizuku	Medium	Shizuku dies on reboot; needs Wi-Fi to restart; shell-level attack surface (Talsec demo)	Sourced
C. Unlock bootloader / root / GSI on this iQOO Z7 Pro	Highest in theory	Unlocked: fails device/strong integrity (hardware-backed)	Poor	Not available	No legitimate unlock found for iQOO Z-series	Sourced (weak)
D. Pixel + GrapheneOS, relocked	High at app level; no system-app privileges for PAA unless you fork	Best: custom-key verified boot, hardware attestation; not Google-certified	Mixed: Paytm reported blocked (Feb 2026); PhonePe, HDFC reported working; Axis/BHIM/GPay unverified	High (new phone, migration)	Fails your Paytm requirement as of reports; sandboxed Play	Sourced (user reports)
E. Pixel 10a (or newer), stock, locked	Medium-high: Android 16+ AppFunctions possible later; same accessibility route as A	Full, Google-certified	Best	Medium (new phone, ₹49,999 launch price in India)	Gemini screen automation is US/Korea only; Android 17 Advanced Protection blocks accessibility if you switch it on	Sourced + Inference

Path	Agent power	Security (verified boot, attestation)	Banking / UPI	Effort	Risks	Evidence
F. Own AOSP fork + PAA as system app, relocked on Pixel	Highest legit	Verified boot with your own key; attestation chain not recognised by Google	Likely poor: not Google-certified, so Play Integrity device/strong will not pass	Very high	You maintain security patches yourself	Inference only
G. Cloud number for AI calls (Exotel/Plivo/Twilio)	Fixes the call gap on any path	n/a	n/a	Medium	TRAI A2P rules; calls come from a business number	Sourced

Path C is blocked on your hardware, so "unlocked plus root" is not a real option for you regardless of preference. Path F is my inference: I found no source showing a self-signed AOSP build passing banking-app checks.

Recommendation and fallback

Recommended: Path A plus G. Stay on the stock locked iQOO, grant PAA its permissions through the normal restricted-settings route, keep Shizuku and wireless debugging off by default, and put AI phone calls on a cloud number. **[Inference from the sources in the next section.]**

1. Scope PAA's accessibility service to the few packages it needs (WhatsApp, the shopping apps) via `android:packageNames`, so it never receives events from bank apps. This reduces what a bank app can object to and what a hostile screen can inject.
2. Read bank data from SMS alerts and bank-app notifications, plus statements you save yourself. Do not automate the Axis app screens.
3. Order food through Swiggy and Zomato MCP servers if your personal client is allowed (Test 7); fall back to accessibility with a hard stop before the payment screen.
4. For calls: cloud number plus realtime voice model, with an AI disclosure line at the start of each call.
5. Treat Shizuku as a one-off provisioning tool, not a running service.

Fallback: Path E, a stock Pixel 10a or newer, only if you want AppFunctions on Android 16+ and longer security patches. It does not fix a bank app that detects accessibility services, because that check runs inside the app on any phone. It keeps Google certification, so banking apps behave as on any certified phone. **[Inference.]**

Not recommended: GrapheneOS (Path D) because Paytm is reported blocked and three of your five payment apps have no solid 2026 report; and an own-AOSP fork (Path F), which costs the most and, by my inference, loses the banking checks you need.

Capability by capability

Most capabilities have a workable legitimate method on Path A; AI speech inside a SIM call, Account Aggregator access and a single API across all merchants do not, and a few items I could not confirm. Links are to the pages I read; dates are publication or access dates where I could see them.

Capability	Best method found	Who has done it	Known limits
Read and schedule tasks from any app	NotificationListener plus AccessibilityService, reading the accessibility tree rather than screenshots	Droidrun reads the accessibility tree and self-reports 91.4% on AndroidWorld (Oct 2025, GPT-5); Roubao is an on-device Kotlin agent	On Android 15, sideloaded apps need "Allow restricted settings" for notification access and accessibility (9to5Google, Sep 2024). Benchmark score is self-reported on an emulator; a harder benchmark, MobileWorld, shows a sharp drop
Send WhatsApp / SMS / email	Accessibility tap for WhatsApp; SMS via granted permission or a pre-filled intent you confirm	Your own PAA already does the WhatsApp part	SEND_SMS and RECEIVE_SMS are "hard-restricted" for sideloaded apps, so the manual "Allow restricted settings" step is needed (forum report, Android 15/16). WhatsApp's guidelines ban unofficial clients and auto-messaging "to harm" WhatsApp or users; they do not address personal-account automation, so ban risk is **[Unverified]**

Capability	Best method found	Who has done it	Known limits
Place calls and wake-up call	Your self-managed ConnectionService and exact alarms	You already built this	No research needed; not a risk to constraints **[Inference]**
AI talking on a normal SIM call	None found on-device. Cloud number with realtime voice agent instead	Indian stacks: Plivo and Exotel as telephony, you bring the AI layer; Bolna as an orchestration layer	Android's own source restricts uplink injection to apps holding MODIFY_PHONE_STATE (AOSP change); that is a system permission, so a normal app cannot use it. Vendor-quoted prices: Plivo India SIP about ₹0.60/min, number about ₹250/month; Exotel packages from ₹9,999 (awaaz.ai, vendor guide, about May 2026)
TRAI and disclosure rules for AI calls	Declare automated calls to your operator or use designated number series; open each call by saying it is an AI	TRAI Third Amendment, 18 Sep 2026: automated and artificial-voice calls are "A2P"; termination charge up to 5 paise/min with exemptions	An on-call disclosure to the callee is only "signalled" per a vendor blog, not confirmed in the regulation text I read **[Unverified]**. Whether a personal assistant calling friends counts as A2P is **[Unverified, ask a telecom lawyer]**. Your own number cannot be used from a cloud provider **[Inference]**

Capability	Best method found	Who has done it	Known limits
Record call audio (not inject)	Shell-level recorder through Shizuku	ShizuCallRecorder lists Android 11-16, Android 15 supported; a commenter reports it working on a Fairphone 4, Android 15, unlocked; scrcpy 3.2 (30 Mar 2025) added voice-call, uplink and downlink sources	I found no list of which phones work; MediaTek/vivo is **[Unverified]**, BCR notes some devices lack the VOICE_CALL source. Recording is useless for talking back
Order food and groceries	Official MCP servers: Swiggy (food, Instamart, Dineout, 27 Jan 2026); Zomato (official open-source server, Sep 2025)	A GitHub demo queries both with OAuth	One X post says neither permits self-serve third-party app use (post); single source, **[Unverified, Test 7]**. Zomato's API policy is old and bars passing data to third parties
Order from Blinkit, Zepto, Amazon, Flipkart, Myntra	Accessibility-driven in the app, stopping before the payment screen	Gemini does this on Pixel 10 / Galaxy S26 for US and Korea food and rideshare apps and requires your confirmation (TechCrunch, 25 Feb 2026)	I found no public ordering API for these five. Hobby scrapers use app endpoints (example); I would not rely on that, as it risks account action. Gemini automation is not offered for your phone or region

Capability	Best method found	Who has done it	Known limits
Compare buses, trains, flights	Read-only browsing of redBus, ixigo and MakeMyTrip with you confirming	None found for an individual	I found no official IRCTC API for individuals. Amadeus Self-Service was decommissioned on 17 Jul 2026, leaving only enterprise access. Scraping terms not checked **[gap]**
Read Axis transactions and find who owes you	Parse bank SMS alerts and notifications; add statements you download	The Gmail-order-parser project shows the parse-and-categorise pattern for email	Axis statement PDF parsing not researched **[gap]**. Account Aggregator is closed to individuals (below)
Bank data via Account Aggregator	None for you as an individual	Sahamati: only entities registered with RBI, SEBI, IRDAI or PFRDA can be FIPs or FIUs; Setu offers sandboxes to such entities	A 2026 overview says a non-regulated startup cannot be an FIU and must partner with a regulated entity. No individual-use route found **[Inference: the "personal" case is not specifically covered]**
Open any app instantly, SOS, web answers	Package launch intents with a foreground service; Gemini API for web	Roubao uses `am start` and `deep_link` tools via Shizuku	Android 15 background-launch limits and OEM battery rules not researched **[gap, Test 6]**; SOS and web Q&A use standard APIs **[Inference]**

Bootloader, custom systems and attestation (questions 1 to 3)

- **Unlock on your phone.** A July 2025 blog lists iQOO Z, U and T series as not unlockable and says 2024-25 phones block downgrades; it is a low-quality source. XDA threads say the developer-menu toggle does nothing on iQOO, and Wikipedia lists Vivo as impossible since May 2022. A GetDroidTips guide names the Z7 Pro among phones for

an EDL method but gives no Z7 Pro result, and a commenter says it failed on an iQOO Z6 in Dec 2023. **I found no confirmed unlock, no custom-key re-lock, and no custom ROM or GSI for the Z7 Pro. [Sourced, absence of evidence only.]**

- **Who can re-lock with custom keys.** GrapheneOS's FAQ lists verified boot with custom key, StrongBox attestation and secure-element throttling as requirements, which Pixels meet; GrapheneOS confirmed the Pixel 10 supports unlock, flash another key and lock. A Motorola partnership targets 2027 per Swappa, second-hand. I did not find current verified-boot details for CalyxOS or /e/OS; the GrapheneOS forum says its Auditor does not rate e/OS and custom builds as secure.
- **Play Integrity.** The device verdict now uses hardware-backed signals and strong requires a security patch within the last year (Android Authority); basic can still pass on unlocked phones. A Telegram post claims many iQOO users fail strong even when stock; **[Unverified, Test 3]**. I found no list of which Indian apps ask for which level.
- **Indian reports on GrapheneOS.** In the GrapheneOS forum thread: Paytm blocks GrapheneOS (Feb 2026), PhonePe works for a user, the new HDFC app failed then worked (30 Jul 2026), and a poster notes that some Indian apps detect GrapheneOS by means other than Play Integrity. A July 2026 guide says UPI works on a Pixel 8a, but it is a secondary blog. Axis, BHIM and Google Pay have entries in the community list but I could not read their status.
- **Rooted or unlocked reports.** I found only workaround guides that hide developer options from banks; those are the spoofing methods you excluded, so I did not rely on them.

Banks versus accessibility, screen capture and sideloading (question 4)

- ICICI documents that its app refuses to run with USB debugging on. A community post says BHIM flagged Shizuku and later Bitwarden's accessibility permission. **Axis: nothing found either way. [Unverified, Test 1.]**
- FLAG_SECURE blocks screenshots and screen recording; in an Appium thread the UI tree stayed readable while screenshots went black. Whether that holds for your accessibility reads on Axis is **[Unverified, Test 2]**; Shizuku screencap on a secure screen should return black **[Inference]**.
- Android 17's Advanced Protection revokes accessibility from non-accessibility apps when switched on. Keep it off on any phone running PAA.
- Google's developer verification is enforced from 30 Sep 2026 in four countries (not India) with global expansion later; the date for India was not found. The "advanced flow" (24-hour wait) rolled out from 18 Aug 2026, and ADB installs are untouched.

Installing PAA with `adb install` avoids the flow.

Shizuku and device owner (question 6)

- Shizuku runs at the ADB shell user (uid 2000), starts via wireless debugging, and must be restarted after every reboot (Flywheel guide, Apr 2026); the thedjchi fork adds boot autostart and a watchdog. It can grant shell-grantable signature permissions, but some are refused on certain OEM builds. Talsec demonstrates a malicious app silently self-granting permissions through it.
- Device-owner mode by `dpm set-device-owner` works only on a phone with no Google accounts or secondary users (Bayton). It adds policy control, not call-audio or injection power **[Inference]**, and on your daily phone it means removing accounts or a factory reset first.

Agent security (question 10)

- **Published attacks.** A July 2026 paper tested five open-source agents and every one fell to at least six of seven attacks: invisible on-screen text, fake overlays that harvest credentials, and accessibility events that expose typed text including passwords. Another paper isolates the accessibility tree as a prompt-injection channel, including hidden metadata. A confirmation prompt did not stop the perception attacks.
- **What I would build (all Inference).** (1) A planner that reads untrusted messages and web text but has no tools, and an executor that accepts only typed, allow-listed actions. (2) Recipient and message body for any send shown from the structured action, not from model text, and approved with BiometricPrompt using a Keystore CryptoObject for first-time contacts and anything money-adjacent. (3) Accessibility limited to named packages and event types, with password and text-change events off. (4) LLM and executor in separate processes; backups off. (5) Use on-device Gemma for anything private and send only the bare web query to Gemini, because anything sent to the Gemini API leaves the phone and conflicts with your constraint 3. StrongBox support on your phone is **[Unverified, Test 9]**.

Added research: bank blocking, terms of service, calls and law (questions 4, 5, 6, 7, 9)

- **Axis and other apps reportedly block on accessibility and developer options.** A May 2026 techenclave thread (link) has users saying Axis's app refused to run on 31 May 2026 while Bitwarden held accessibility permission, and one user infers any enabled accessibility app triggers it (Tasker and Nova were also mentioned). Apps reported

refusing with developer options on: RailOne, DigiLocker, SBI YONO, the new Aadhaar app, mParivahan. HDFC and ICICI reportedly moved from block to warning; IndusInd and HSBC blocks were reportedly reversed after complaints. These are forum comments, not verified reporting. The thread does not mention BHIM, Google Pay, PhonePe, Paytm or Shizuku. A technofino post (link) says BHIM flagged Shizuku and Bitwarden accessibility, and users run a Tasker profile that switches those off when BHIM opens. **[Sourced as user reports; Unverified for your phone.]**

- **Accessibility gating idea.** PAA could keep its accessibility service off by default and switch it on only for non-banking apps, using a one-time ADB grant of WRITE_SECURE_SETTINGS plus foreground-app detection. This disables rather than hides the service, so it is not spoofing. **[Inference, untested: Test 11.]** Keep developer options off in daily use and re-enable only for ADB steps.
- **Axis e-statement PDFs.** The password is the first four letters of your name in uppercase plus DDMM of your birth date, or plus the 9-digit customer ID; name must match the account (ClearTax). Once unlocked, PAA can parse the PDF locally. **[Sourced for the format; parsing is Inference.]**
- **WhatsApp terms.** The terms (link) prohibit non-personal use unless authorized, bulk messaging, auto-messaging and auto-dialing, and reverse engineering. Auto-tapping Send on your own chats is a gray area, so account-ban risk is real. **[Sourced text; enforcement Unverified.]**
- **Shopping and rail terms.** Flipkart (terms) bars robots, spiders, page-scraping and deep-linking by automated methods. A copy of Amazon's conditions bars data mining and robots. IRCTC (terms PDF) says use of automation or scripting software is strictly prohibited, with possible deactivation and ticket release without refund. I found no official individual IRCTC API, only agent resellers. Use IRCTC for reading and planning by hand, not for automated booking. I could not read Myntra, Blinkit, Zepto or amazon.in terms.
- **Relocking other systems.** CalyxOS says relocking is safe on Pixels (doc); /e/OS reportedly cannot relock a Pixel 7 (thread, old).
- **Call audio.** scrcpy 3.2 lists voice-call, voice-call-uplink and voice-call-downlink sources, working out of the box on Android 12+ (doc); no list of phones where they work on MediaTek or vivo exists, so Test 4 decides. Separately, an iQOO forum post (link, 2023) says setting vivo's own Phone app as default enables built-in call recording; whether both sides are captured is not stated. **[Unverified.]**

- **Law.** Indian law does not explicitly require consent for recording your own calls, but privacy and evidence rules make two-party consent the safer course; DPDP treats recordings and transcripts as personal data, and a mandatory AI-caller disclosure is argued for but not yet a hard rule. This is a summary of law-firm and vendor articles, not legal advice.
- **Other agents.** Tasker AutoInput relies on accessibility and users report Android 15 killing the service repeatedly; Rabbit R1 runs on Android and marketed a "large action model"; Humane's AI Pin service ended in February 2025 after HP bought its assets. The Rabbit and Humane details come from summaries, not full reads.

Solutions other people found, problem by problem (third round, 16 Oct 2026)

Honest note on method: my search tools returned almost no Reddit threads (only an r/tasker mirror on an unrelated battery issue), so the solutions below come from project pages, vendor docs and forums. Where a problem has no found solution I say so.

- **AI talking on a normal SIM call, no root, no spoofing: Bluetooth hands-free from a second device.** Sipfront runs a Raspberry Pi as a Bluetooth Hands-Free Profile device (like a car kit) using oFono and PipeWire; it can dial, hang up and inject and record call audio, and works with any paired HFP phone (Sipfront, 20 Mar 2025, updated 2 Jul 2025). ParrotVoice (launched 2026) does the same from a Windows 11 PC or an ESP32 board with nothing installed on the phone (Product Hunt). Limits: Bluetooth audio is 8 or 16 kHz with added delay, the computer or board must stay on and near the phone, and the AI runs on that device, not on the phone. The rooted alternative, AgentCall, needs Magisk and has only been qualified on one Poco phone (GitHub), so it fails your no-root rule. With local speech-to-text and text-to-speech on that device, the call audio need not leave your home (**[Inference]**). Whether your iQOO exposes full call audio over HFP is **[Unverified: Test 12]**. A cloud number (Path G) remains the option when you are away from home.
- **Ordering food: use the official agent routes, not screen-tapping.** Swiggy opened Builders Club in April 2026 with Food, Instamart and Dineout MCP servers; its site lists personal use as "connect your AI client, no access required" via OAuth, while production apps need review (Swiggy Builders). UPI checkout through GPay, PhonePe, Paytm or QR was added 10 Jul 2026 (Swiggy). A Swiggy Money wallet route (4 Sep 2026) lets an agent pay from a pre-loaded balance without a per-order approval; confirmation is only a model instruction, not enforced, so PAA must force its own confirm and keep the wallet balance small (Stellagent, 8 Sep 2026). An April 2026 write-up found v1 was COD-only with a Rs 1,000 order cap and that retries after a

server error are unsafe (Khare, 30 Apr 2026). Zomato's MCP server supports search, cart and checkout with a UPI QR you scan in your bank app, but only allow-listed redirect URLs work (Claude Desktop, ChatGPT) (Analytics Vidhya, 11 Nov 2025). Reuters is reported to say NPCI is preparing a Unified Agent Protocol for agentic UPI payments, not final. This removes most need for accessibility on Swiggy and Zomato, which helps the bank-app problem. Whether a custom Android app can complete their OAuth redirect is **[Unverified: Test 13]**. For Blinkit, Zepto, Amazon, Flipkart and Myntra I found no official agent route, so those still need on-screen driving with your fingerprint and PIN at the end, plus the terms-of-service risk above.

- **Trains, buses, flights.** ixigo put Flights, AbhiBus (buses) and ConfirmTkt (trains) into ChatGPT on 29 Apr 2026 (Motor India); I found no public API or MCP server for a personal agent, and IRCTC prohibits automation. Practical answer: let PAA open ixigo, redBus and MakeMyTrip pages for you, or ask ChatGPT's ixigo app, and compare by reading the screens yourself.
- **Bank transactions.** No individual route through Account Aggregator was found: Sahamati's page describes sandboxes for teams and says nothing about individuals (Sahamati FAQ), consistent with my earlier finding that only regulated entities can be data users. The workable route is on-device SMS parsing. PennyWise AI is an open-source (AGPL-3.0) F-Droid app that reads bank and UPI SMS on the phone, claims 160 banks (Axis is not named), and had version 2.22.0 on 15 Oct 2026 (F-Droid). It has an optional webhook sync that can send data out; leave it off. Reuse its parsers or run it beside PAA, and test it on your own Axis alerts (**[Test 14]**). Axis statement PDFs add detail (password format above).
- **Accessibility policy risk.** Android's Advanced Protection mode revokes accessibility for apps not marked as accessibility tools; it is opt-in and was found in Canary 2602, not officially announced (Android Authority, 13 Feb 2026). A March 2026 report says Android 17.2 tightens the rules and applies the block when that mode is on (Security Boulevard, 17 Mar 2026); whether sideloaded apps are covered otherwise is not stated. Do not enable Advanced Protection on the phone that runs PAA.
- **Bootloader on your phone: no solution exists.** iQOO users have asked for unlocking since at least Dec 2023 with no official answer visible (iQOO community). The only unofficial vivo unlock I found works on a few Snapdragon models (V21e, X70 Pro+, Y31) via a race condition and breaks the fingerprint reader on one (XDA); no Dimensity phone is listed, and an unlocked bootloader would break your first constraint anyway.
- **Axis blocking PAA: no found fix beyond switching the service off.** The only user workaround found is a Tasker profile that turns Shizuku and accessibility off when the bank app opens (technofino post above). One further idea, **[Inference, untested: Test**

15]: keep banking apps in a separate Android user or work profile where PAA's accessibility service is not enabled, since accessibility settings are stored per user; I did not confirm this with a source. If Test 11 and Test 15 both fail, the honest answer is that Axis and a phone-wide accessibility agent cannot share a phone without spoofing, which you excluded; then run PAA's screen-driving on a second phone with no banking apps, which keeps all your constraints on the main phone **[Inference]**.

The two hardest problems, researched everywhere (fourth round, 16 Oct 2026)

Reddit pages were blocked or unreadable to my tools (one reader mirror returned only a bot check), so I relied on security-vendor write-ups, news and forums, including reports that quote Reddit users.

Problem 1: Axis refusing to run next to an accessibility agent.

- **How apps decide.** Banking apps list enabled services with `AccessibilityManager` and either block on any service (a presence check) or check each against a trusted package list; the author of one guide says a legitimate user must disable the service for that app, and that banks may add a service to the list only if many users complain (Wultra, Guardsquare). Malware heuristics also score a non-system app that reads window content, listens to window-change events, asks for overlay permission or was installed minutes ago (devsec-blog, Mar 2024). PAA would trip several. Which method Axis uses is **[Unverified]**.
- **Install source may matter.** In January 2026 HSBC UK's app locked out users who had installed Bitwarden from F-Droid. Bitwarden said the app can see whether other apps came from Google Play; the exact trigger is unconfirmed, and HSBC gave no fix (The Register, 7 Jan 2026). Commenters claimed the block hits any non-Play app that can read the screen (Born's Tech, 4 Jan 2026); that is unverified. This is a UK bank, not Axis. **[Sourced as reports; applies to Axis only by analogy.]**
- **Fixes others suggested.** F-Droid's Neil Brown suggested banking in a separate device profile or on a separate device, and now uses the bank's website; a commenter said an alternate Android profile may suffice for some HSBC regional apps. A 2024 Reddit user, quoted by Born's Tech, was offered the choice to disable accessibility for the app or uninstall the password manager. This supports Test 15 and the second-phone option, but no source shows it working for Axis.
- **Dead end: Google Play.** Play's policy says automation tools and assistants do not qualify as accessibility tools, allows only deterministic, human-defined scripts for other apps, and prohibits autonomous planning and execution (Play Console Help).

Publishing PAA to Play to get a Play install source is therefore not a legitimate route for an autonomous agent. Declaring `isAccessibilityTool` falsely would be misrepresentation, so I do not suggest it.

- **Ranked options.** (1) Gate the service off around bank apps (Test 11). (2) Keep banking in a separate profile (Test 15). (3) Ask Axis support directly whether third-party accessibility services are blocked and whether they allow-list any (Test 16). (4) Run PAA's screen-driving on a second phone with no banking apps. (5) Use the official Swiggy and Zomato routes and notification replies so fewer tasks need the accessibility service at all. None is proven.

Problem 2: bootloader on the iQOO Z7 Pro. Nothing found changes the earlier answer: there is no official unlock, and iQOO users have asked since Dec 2023 (iQOO community). A phone-unlocking watchlist puts vivo and iQOO in its "avoid at all costs" tier (wall of shame), and the one unofficial vivo method covers only a few Snapdragon models (XDA, above). Phones users report as relockable with their own keys, per the avbroot tracker (issue 299, user reports): Pixel, Fairphone 3 to 5, Nothing Phone (1), (2), (3), (3a) and (4a), several Motorola moto g models, some Sony Xperia models, OnePlus 6 and 6T; Xiaomi and Samsung are listed as not supported, and there are no MediaTek or vivo examples. Motorola and GrapheneOS had no finished support list as of March 2026 (DEV, low-quality source). Whether Axis and UPI apps accept a custom-key relocked non-Pixel is **[Unverified]**. The legitimate answer for this phone: keep it stock and locked, and run PAA as an ordinary unprivileged app. The bootloader question only matters if you change phones, in which case the Pixel path in the table remains the best documented.

Tests to run on your phone

Run Tests 1, 3 and 10 first: they decide whether Path A survives or you move to the Pixel fallback. Do each on a fresh boot and note the app versions.

1. **Do banks object to PAA's accessibility service?** Enable PAA's service. Open Axis Mobile, BHIM, Google Pay, PhonePe and Paytm, log in and open the payments screen. Record any warning, forced logout or block, then repeat with the service off. Then scope PAA with `android:packageNames` to WhatsApp only and repeat; the question is whether a bank checks for any enabled third-party service or only for services that watch it.
2. **Can PAA read Axis screens under FLAG_SECURE?** With the service on, open the Axis transactions page and dump `AccessibilityNodeInfo` text from PAA for that window. Compare with a screenshot. Run it only against your own account and do not automate

taps in this app.

3. **Which Play Integrity verdicts does your phone return?** Install a Play Integrity checker app from the Play Store and note basic, device and strong. Repeat after the latest monthly patch. If strong fails on a stock locked phone, note it: that confirms the report about iQOO phones and tells you which banking apps are at risk.
4. **Does shell-level call recording work on your chipset?** On a PC with scrcpy 3.2 or newer and USB debugging on, place a normal SIM call and run `scrcpy --no-video --no-playback --audio-source=voice-call --record=call.opus`; repeat with `voice-call-downlink` and `voice-call-uplink`. Silence on the far end means the modem path is not exposed. Only the shell route is tested here; PAA gains nothing without it.
5. **SMS parsing on Android 15.** Install PAA by `adb install`, open Settings > Apps > PAA > three-dot menu > Allow restricted settings, grant SMS and notification access, then have Axis send a small UPI debit alert. Check PAA sees it and parses amount, merchant and balance.
6. **Instant app launch from the wake word.** With the screen locked and then with another app in front, trigger PAA's wake word and have it open a banking app and WhatsApp. Note which launches Android 15 blocks without a full-screen intent or overlay permission.
7. **Swiggy and Zomato MCP.** Read the setup pages in Swiggy's and Zomato's GitHub repos; register a client with OAuth from a small script, search a menu, build a cart and stop before checkout. If registration is refused, note the stated terms; that decides whether the accessibility route is needed.
8. **Cloud-number AI call.** Trial an Exotel or Plivo number, connect it to a realtime voice model, call your own second phone, open with an AI disclosure, and measure the delay from your speech ending to the reply. Check what number the receiver sees and whether your contacts' phones flag it as spam.
9. **StrongBox and key attestation.** Run `adb shell pm list features | grep -i strongbox`. If absent, plan Keystore keys in the TEE only.
10. **What triggers the bank warnings.** On a clean boot, test each bank app with: Developer options off; Developer options on; wireless debugging on; Shizuku installed but stopped; Shizuku running. This tells you whether a rarely-run Shizuku session is safe or whether you should avoid it entirely.

Test 11 (added): gate your accessibility service around bank apps. (a) With developer options off and PAA's accessibility service on, open Axis, BHIM, Google Pay, PhonePe and Paytm one by one and note whether each runs, warns or refuses. (b) Turn PAA's service off and repeat. (c) If only (a) fails, grant PAA WRITE_SECURE_SETTINGS once over ADB

(`adb shell pm grant <your.package> android.permission.WRITE_SECURE_SETTINGS`), then have PAA turn its own service off when a bank app is in the foreground (UsageStats) and on again afterwards. (d) Also check whether the built-in vivo Phone app's call recorder captures both sides (Test 4). If (c) fails or lags, Path A is not viable for Axis and you should accept manual banking with PAA off. **[Inference, untested.]**

Test 12 (added): AI on a SIM call over Bluetooth. Pair a Raspberry Pi (oFono and PipeWire, as in the Sipfront article) or a Windows 11 PC with ParrotVoice to the iQOO as a hands-free device. Place a call to a friend; play a test phrase into the call and record what comes back. Pass if both directions are clear and delay is tolerable.

Test 13 (added): Swiggy and Zomato from PAA. On localhost, add Swiggy's MCP server to an MCP client (Swiggy says no access is needed for personal use) and run one cart build, stopping before payment. Then try the same OAuth from a small Android client; if the redirect is refused, note the error and use a laptop client instead.

Test 14 (added): SMS parsing. Install PennyWise AI from F-Droid with webhook sync off, grant SMS, and check that your last 30 Axis debit, credit and UPI alerts are categorised correctly. If not, copy the failing messages into PAA's own parser tests.

Test 15 (added): profile separation. Create a work profile or secondary user, install Axis there, and keep PAA's accessibility service enabled only in the main profile. Open Axis; pass if it runs without a block.

Test 16 (added): ask Axis. Message Axis Bank support from the app or a branch: "Does the Axis Mobile app block when a third-party accessibility service is enabled or developer options are on, and is there an allow-list?" Keep the written reply; it settles Axis's rule better than any forum. **[Inference that they will answer.]**

If Tests 1 and 10 pass and Test 3 returns device-level or better, stay on Path A. If Axis, BHIM or Google Pay block the service in Test 1, a new phone will not help: use the on-off gating in Test 11 or the two-phone split described in the added research below.

What is proven, what is inferred, what I did not cover

The core platform facts rest on official or first-party pages; the banking and unlock facts rest mostly on user reports and blogs, so treat them as strong hints, not guarantees.

Proven by first-party or official sources

- GrapheneOS requires custom-key verified boot, StrongBox attestation and a secure element, which only Pixels meet today (FAQ).

- Android 15 restricts notification listening, accessibility and SMS for sideloaded apps unless you allow restricted settings (9to5Google).
- AppFunctions needs Android 16 and its Gemini integration was in private preview as of May 2026 (Android developer docs), so it is not usable on your Android 15.
- Swiggy and Zomato run official MCP servers; Amadeus Self-Service closed on 17 Jul 2026; TRAI amended its anti-spam rules on 18 Sep 2026 to cover automated and artificial-voice calls.
- Account Aggregator participants must be entities regulated by RBI, SEBI, IRDAI or PFRDA (Sahamati).

Supported only by user reports, blogs or vendor pages

- iQOO phones cannot be unlocked; Paytm blocks GrapheneOS; BHIM flags Shizuku and some accessibility apps; vendor prices for Plivo and Exotel; "many iQOO users fail strong integrity" (Telegram post, weakest).

My inference, not found in any source

- Path A satisfies all four constraints; a self-signed AOSP fork loses banking checks; scoping accessibility to named packages reduces bank objections; a cloud number cannot show your own caller ID; device-owner mode adds no call-audio power; Gemini calls conflict with keeping private data on the phone.

Not covered or not confirmed

- Whether Axis, BHIM, Google Pay, PhonePe and Paytm block your specific accessibility service on your phone (only user reports exist); which Indian apps use which Play Integrity level; which vivo or MediaTek phones expose voice-call audio to the shell.
- An official IRCTC access route for individuals (none found); the automation terms of Blinkit, Zepto, Myntra and amazon.in (pages unreadable to me); how WhatsApp enforces its terms against personal-account automation.
- Primary-source detail on Rabbit and Humane (summaries only); a lawyer's view on recording calls and on an AI speaking for you under DPDP (I read secondary articles only); whether India's developer-verification date for sideloaded apps has been set.
- No Reddit thread was retrievable with my tools, so community evidence is thinner than you asked for. No tested fix for Axis blocking an accessibility agent was found; per-profile separation is untested and unsourced. Whether Swiggy's individual-developer FAQ allows a personal Android client, and whether a custom app can pass Zomato's redirect allow-list, is unknown.

- Axis's actual detection logic (presence check, allow-list, install source or heuristics) is not public; only Test 11, 15 and 16 can settle it. No vendor or forum source shows a coexistence method for a screen-reading agent and Axis.

Sources

Research run on 16 Oct 2026. I opened three pages in full (Amadeus, WhatsApp guidelines, Appium thread); the rest I read as search excerpts, so check any link before you rely on a detail.

- Bootloader unlocking, Wikipedia (2026)
- Vivo/iQOO unlock blog (Jul 2025, low quality); XDA thread; GetDroidTips (Apr 2025 comments)
- GrapheneOS FAQ; GrapheneOS Pixel 10 post; Swappa on GrapheneOS (2026); Auditor discussion
- GrapheneOS India thread (Feb to Jul 2026); PrivSec banking list; Vucense India guide (Sep 2026)
- Android Authority on Play Integrity; Tadi Telegram post
- Android 15 sideload limits, 9to5Google (12 Sep 2024); SMS restricted-settings thread
- Android developer verification, Google; Sep 2026 enforcement summary; Android 17 Advanced Protection
- ICICI USB debugging notice; BHIM and Shizuku thread; Appium FLAG_SECURE thread
- Sahamati: joining AA; AA 2026 overview; Finvu
- Flywheel Shizuku guide (Apr 2026); ShizuCallRecorder; scrcpy 3.2 news (2 Apr 2025); Bayton on device owner; Talsec on Shizuku
- Telephony stack guide, awaaz.ai (2026); TRAI rules, Telecom Lead (Sep 2026); Telangana Today, 18 Sep 2026; VoiceLink on TRAI series
- AppFunctions docs; 9to5Google AppFunctions (25 Feb 2026); Gemini Intelligence, Engadget (12 May 2026); TechCrunch (25 Feb 2026)
- Droidrun benchmark; MobileWorld; Roubao; AutoGLM paper
- Swiggy MCP, Inc42 (27 Jan 2026); Zomato and Swiggy MCP, Tru Commerce (25 Jul 2026); Zomato API policy
- Amadeus shutdown, PhocusWire (9 Feb 2026); WhatsApp messaging guidelines
- Mobile-agent attacks, The Hacker News (Jul 2026); arXiv 2607.00333; arXiv 2608.08939
- Pixel 10a India launch (18 Feb 2026)

- techenclave: apps failing with developer options (May 2026)
- technofino: BHIM flagging Shizuku and accessibility
- ClearTax: Axis statement password
- WhatsApp Terms of Service
- Flipkart Terms of Use
- IRCTC Terms and Conditions (PDF)
- CalyxOS bootloader lock guide
- /e/OS community: relock Pixel 7
- scrcpy audio documentation
- iQOO community: call recording on Android 12+
- caller.digital: DPDP and AI calling in India (vendor blog, 2026)
- Tasker AutoInput accessibility thread
- Sipfront: automating calls with Bluetooth HFP on a Raspberry Pi (2025)
- ParrotVoice on Product Hunt (2026)
- AgentCall (GitHub)
- Swiggy Builders
- Swiggy MCP UPI payments (10 Jul 2026)
- Stellagent: Swiggy Money in MCP (8 Sep 2026)
- Khare: Swiggy party agent (30 Apr 2026)
- Analytics Vidhya: Zomato MCP (11 Nov 2025)
- Motor India: ixigo in ChatGPT (29 Apr 2026)
- Sahamati: AA hackathon FAQ
- PennyWise AI on F-Droid
- Android Authority: Advanced Protection and accessibility (13 Feb 2026)
- Security Boulevard: Google accessibility crackdown (17 Mar 2026)
- iQOO community: bootloader unlock request (2023)
- XDA: unofficial vivo bootloader unlock
- Wultra: blocking untrusted accessibility services
- Guardsquare: accessibility threats
- devsec-blog: detecting banker malware (Mar 2024)
- The Register: HSBC and sideloaded Bitwarden (7 Jan 2026)
- Born's Tech: HSBC app and Bitwarden (4 Jan 2026)

- Google Play: AccessibilityService policy
- avbroot supported devices (issue 299)
- Bootloader unlock wall of shame
- DEV: Motorola, GrapheneOS and relock